CHAITANYA PANDEY

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in AI systems, backend engineering, and developer tooling. Built production-style systems involving Retrieval-Augmented Generation (RAG), local LLM inference, computer vision, AST parsing, distributed scheduling, and microservices architectures using FastAPI, Spring Boot, React, PyTorch, TensorFlow, and vector databases. Internship experience in database optimization, real-time analytics, and enterprise-scale backend systems.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++

Backend: FastAPI, Spring Boot, Flask, Express.js, REST APIs, Microservices

Frontend: React, Next.js, D3.js, Tailwind CSS, Electron

AI/ML: LLMs, RAG Systems, PyTorch, TensorFlow, FAISS, ChromaDB, Sentence-Transformers, Computer Vision

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Microsoft SQL Server, Amazon RDS

Tools: Git, Maven, Postman, Tree-sitter, NetworkX

Concepts: System Design, AST Parsing, Dependency Graphs, Secure Authentication, Distributed Scheduling

EXPERIENCE

Tata Motors Limited — Engineering Intern (Digital Safety) | May 2025 – June 2025

- Built a real-time safety analytics dashboard aggregating and visualizing operational incident data for internal monitoring workflows.
- Automated ingestion and reporting pipelines, reducing manual reporting overhead and improving traceability across multi-source safety logs.
- Integrated centralized data processing workflows to improve incident auditing and reporting consistency.

Scalability Engineering — Database Engineering Intern | June 2024 – July 2024

- Optimized Microsoft SQL Server queries through indexing and query restructuring, reducing execution time by approximately 40%.
- Redesigned database schemas to improve scalability, consistency, and production data integrity.
- Worked on enterprise-scale database operations, query tuning, and backend performance optimization.

PROJECTS

Code Atlas — AI-Powered Codebase Architecture Visualizer

Tech: FastAPI, React, D3.js, Tree-sitter, NetworkX, FAISS, Llama.cpp

- Engineered a multi-language parsing engine using Tree-sitter and NetworkX to construct AST-driven dependency graphs for large repositories.
- Developed automated frontend-to-backend API linkage detection across JavaScript/TypeScript and Python services.
- Implemented a hybrid RAG pipeline using FAISS and sentence-transformers to enable semantic code search and context-aware AI reasoning.
  Deployment Link: - https://code-atlas-eight.vercel.app/

EL-GAN — Low-Light Image Enhancement GAN

Tech: Python, PyTorch, Torchvision, rawpy

- Engineered a lightweight GAN for low-light enhancement achieving PSNR 18.56 and SSIM 0.74 on the LOL benchmark.
- Built a RAW image processing pipeline supporting high-exposure amplification and domain adaptation for near-zero visibility scenes.
- Optimized a multi-stage adversarial training pipeline combining perceptual refinement and specialized loss functions for texture preservation.

Nutri-Verify — AI Food Label Forensics Engine

Tech: Python, Streamlit, ChromaDB, Sentence-Transformers, Gemini API, Llama.cpp

- Developed a multimodal AI pipeline combining computer vision, RAG, and local LLM inference to audit food labels against FSSAI standards.
- Implemented domain-specific retrieval systems using ChromaDB and sentence-transformers across regulatory and nutritional datasets.
- Built hallucination-constrained report generation workflows using local Llama models for deterministic compliance analysis.
  Deployment Link: - https://chaitu1712-nutri-verify.hf.space/

AI-QKD Privacy Simulator

Tech: Python, TensorFlow, Scikit-Learn, Diffprivlib

- Implemented the BB84 Quantum Key Distribution protocol with noise modeling and intercept-resend attack detection.
- Developed TensorFlow-based predictive models to estimate channel reliability and optimize reconciliation workflows.
- Integrated differential privacy mechanisms using Diffprivlib to secure simulation analytics and operational metadata.

Code Converter — LLM-Based Code Translation Platform

Tech: Spring Boot, Flask, Next.js, Gemini API

- Architected a decoupled microservices system enabling bidirectional Java ↔ Python code translation workflows.
- Integrated Gemini-based inference pipelines for syntactically accurate code conversion and contextual explanations.
- Developed a responsive Next.js frontend with optimized API communication and syntax visualization.
  Deployment Link: - https://code-converter-dun.vercel.app/

EDUCATION

Vellore Institute of Technology (VIT), Vellore — B.Tech in Computer Science & Engineering | 2022 – 2026

CGPA: 7.9 / 10